

CHEMICAL KINETICS

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Chemical Kinetics

The branch of physical chemistry which deals with the rate of reactions. The study of chemical kinetics includes

- The rate of reactions & rate law.
- The factors which affect the rate of reactions

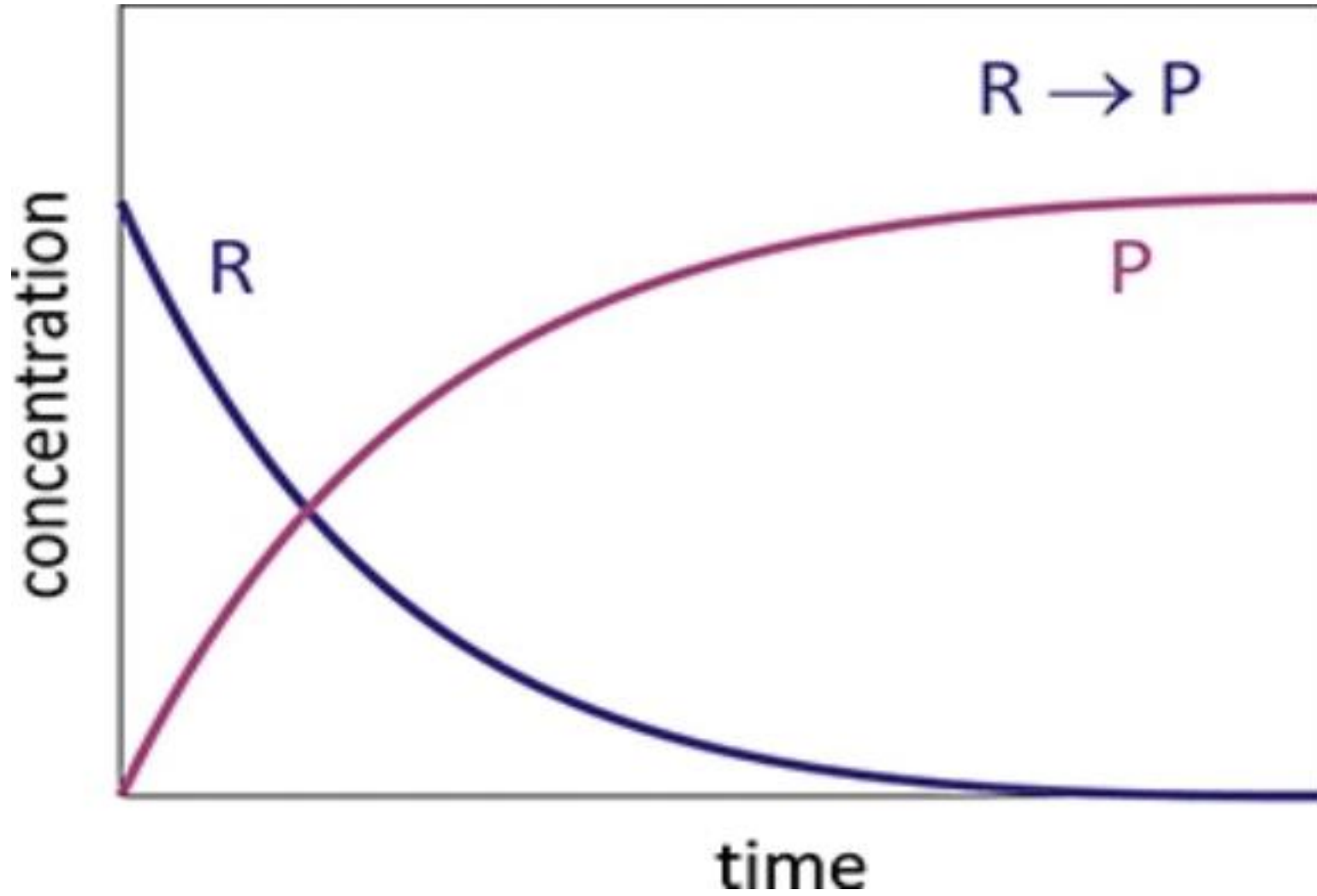
Reaction Rate

The change in concentration of any of reactant or products per unit time. Let us consider a simple reaction $A \rightarrow B$

The concentration of the reactant A decreases and that of B increases as time passes.

rate of reaction = $-d[A]/dt$, rate of disappearance of A
= $+d[B]/dt$, rate of appearance of B

Fig : Reaction Rate

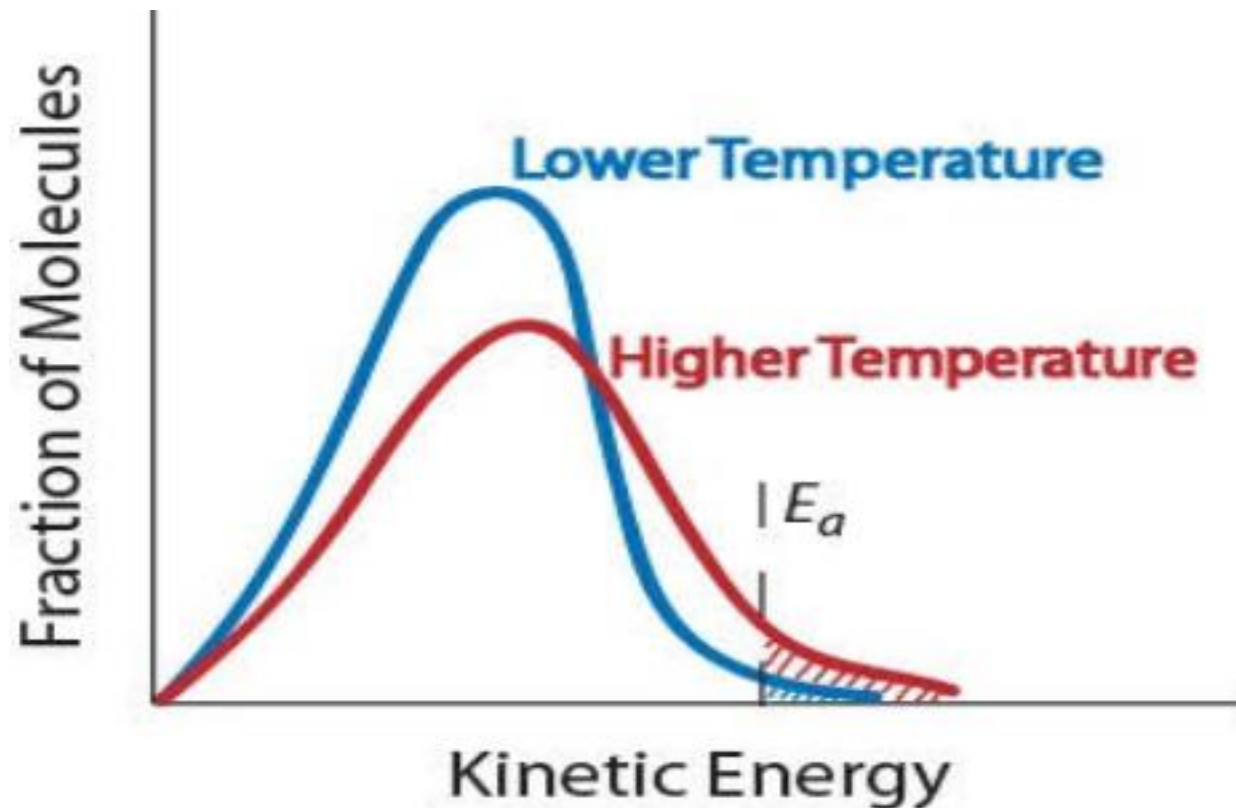


Factors affecting rate of a chemical reaction

- Temperature
- Concentration
- Pressure
- Surface area
- Presence of catalyst

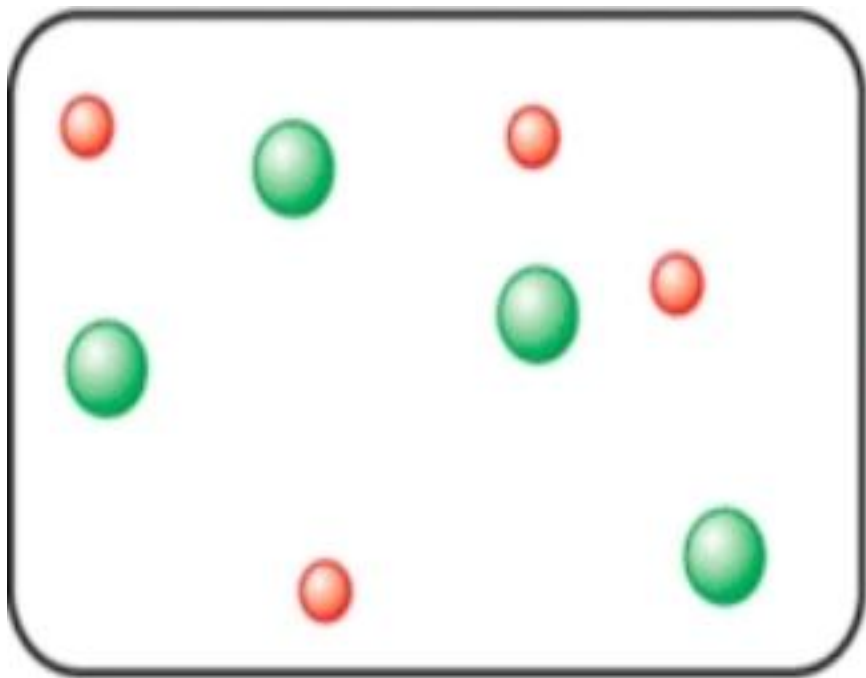
FACTOR 1: Temperature

Increase in temp. $\rightarrow$ increase in K.E. $\rightarrow$ increase in no. collisions
+ increase in no. of particles with greater than required amount of
activation energy $\rightarrow$ more particles react $\rightarrow$ increase rate of reaction.



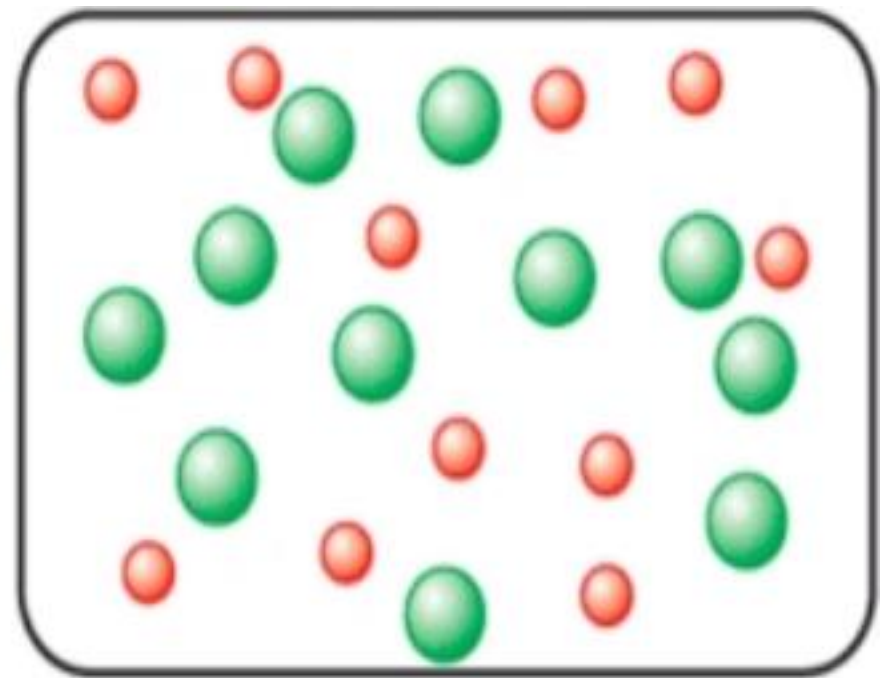
FACTOR 2: Concentration/ Pressure

High concentration/pressure $\rightarrow$ more particles per unit volume $\rightarrow$ Increase in frequency of collisions $\rightarrow$ rate of reaction increases.



Here we have a few molecules.
There are few collisions.
The rate of reaction is low.

Increase
concentration



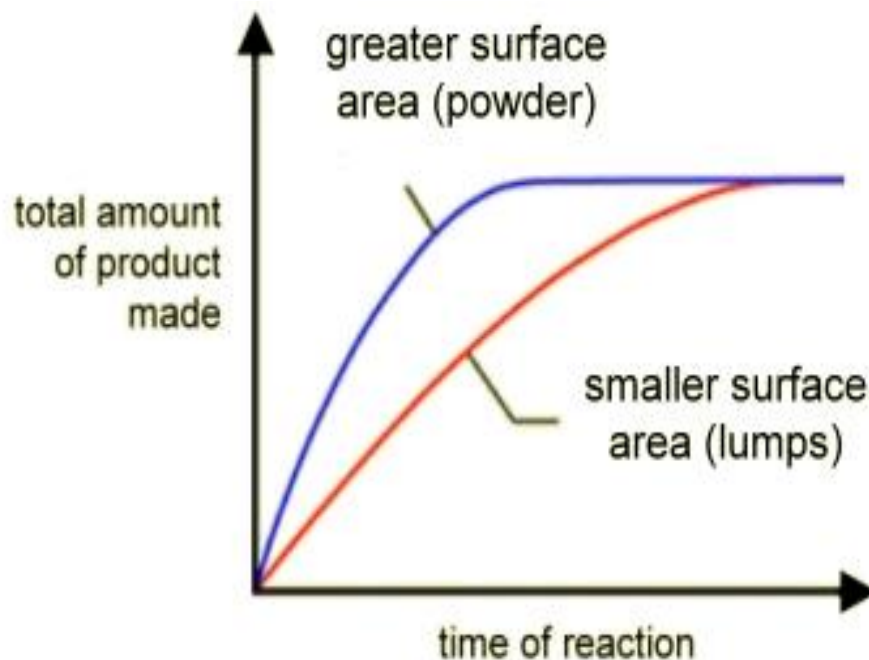
Here we have many molecules.
There are more collisions.
The rate of reaction is greater.

FACTOR 3: Surface Area

Increase in surface area / particle size → increase in exposure to the other reactant → increase in probability of collisions → increase in rate of reactions

Calcium carbonate and hydrochloric acid

In the lab, powdered calcium carbonate reacts much faster with dilute hydrochloric acid than if the same mass was present as lumps of marble or limestone.



Rate Laws

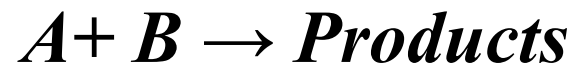
The rate of a reaction is directly proportional to the reactant concentrations, each concentration being raised to some power.

Rate Law For Equation With One Reactant



$$\text{Rate} = k [A]^m$$

Rate Law For Equation With Two Reactant



$$\text{Rate} = k [A]^m [B]^n$$

‘ k ’ is the law constant; Temperature Related

‘ m ’ and ‘ n ’ are rate law exponents

Order of reaction & Molecularity

- **Order of reaction**

The sum of the powers of concentrations in the rate law.

- **Molecularity**

The number of reactant molecules involved in a reaction.

Molecularity	Elementary Reaction	Rate Law
<i>Unimolecular</i>	$A \longrightarrow \text{products}$	$\text{Rate} = k[A]$
<i>Bimolecular</i>	$A + A \longrightarrow \text{products}$	$\text{Rate} = k[A]^2$
<i>Bimolecular</i>	$A + B \longrightarrow \text{products}$	$\text{Rate} = k[A][B]$
<i>Termolecular</i>	$A + A + A \longrightarrow \text{products}$	$\text{Rate} = k[A]^3$
<i>Termolecular</i>	$A + A + B \longrightarrow \text{products}$	$\text{Rate} = k[A]^2[B]$
<i>Termolecular</i>	$A + B + C \longrightarrow \text{products}$	$\text{Rate} = k[A][B][C]$

Difference between Order of reaction and Molecularity

Order of reaction

1. It is determined experimentally.
2. It can have fractional value.
3. It can assume zero value.
4. It can change with the conditions of reaction such as temp., pressure, conc.

Molecularity

1. It is a theoretical concept.
2. It is always a whole number.
3. It can not have zero value.
4. It is invariant for a chemical equation.

Zero Order Reactions

In a zero order reaction, rate is independent of the concentration of the reactions. Let us consider a zero-order reaction of the type



$$\text{At } t=0 \quad a \quad 0$$

$$\text{At } t=t \quad (a-x) \quad x$$

We know that for a zero order reaction, the rate of reaction, dx/dt , is directly proportional to the concentration of the reactant. Thus,

$$dx/dt = k(a-x)^0$$

$$\text{or, } dx = kdt$$

$$\text{or, } \int dx = k \int dt$$

$$\text{or, } x = kt + I \dots\dots (1)$$

where I is the constant of integration. The constant k may be evaluated by putting $t = 0$ and $x = 0$.

Thus, $I = 0$

$$x = kt \text{ or, } k = x/t$$

First Order Reaction

When the reaction rate depends on the first power of concentration of a single reactant it is considered to be first order reaction.

Let us consider a first order reaction



$$\text{At } t=0 \quad a \quad 0$$

$$\text{At } t=t \quad (a-x) \quad x$$

We know that for a first order reaction, the rate of reaction, dx/dt , is directly proportional to the concentration of the reactant. Thus,

$$dx/dt = k(a-x)$$

$$\text{or, } dx/(a-x) = kdt$$

$$\text{or, } \int dx/(a-x) = k \int dt$$

$$\text{Or, } -\ln(a-x) = kt + I \dots\dots (1)$$

where I is the constant of integration. The constant k may be evaluated by putting $t = 0$ and $x = 0$.

$$\text{Thus, } I = -\ln a$$

Substituting for I in equation (1)

$$\ln \frac{a}{a-x} = kt$$

$$k = \frac{1}{t} \ln \frac{a}{a-x}$$

Change $k = \frac{2.303}{t} \log \frac{a}{a-x}$

The characteristics of first order reaction

1) A first order reaction is never complete.

For a first order reaction we have $[A] = [A]_0 e^{-kt}$ (1)

Where $[A]_0$ is the concentration of the reactant $t = 0$ and

$[A]$ is the concentration of the reactant $t = t$

When the reaction reaches completion, the value of $[A]$ is zero, putting $[A] = 0$ into equation (1), we get

$$0 = [A]_0 e^{-kt}$$

$$\text{Or, } e^{-kt} = 0 \quad [\because [A]_0 \neq 0]$$

But $e^{-kt} = 0$ only when $t = \infty$. This means a first order reaction takes infinite time for its completion.

2) The half life of a first order reaction is independent of its initial concentration.

The half - time of a reaction is defined as the time required to reduce the concentration of the reactant to half of its initial value.

When $x = a/2$, $t = t_{1/2}$

Putting these values in following reaction

$$k = \frac{2.303}{t_{1/2}} \log \frac{a}{a - \frac{a}{2}} = \frac{2.303}{t_{1/2}} \log 2 = \frac{2.303}{t_{1/2}} \times 0.30103$$

$$(\because \log 2 = 0.30103)$$

$$t_{1/2} = \frac{0.693}{k}$$

Second Order Reaction

The rate of reactions depends on molar concentration of two reactants or second power of one reactant involved in the reaction



Suppose the initial concentration of A is a moles litre⁻¹. If after time t , x moles of A have reacted, the concentration of A is $(a - x)$. We know that for such a second order reaction, rate of reaction is proportional to the square of the concentration of the reactant. Thus,

$$dx/dt = k(a-x)^2 \quad \dots (1)$$

$$dx/(a-x)^2 = kdt \quad \dots (2)$$

On integration, it gives

$$1/a-x = kt + I \quad \dots (3)$$

where I is integration constant. I can be evaluated by putting $x = 0$ and $t = 0$. Thus,

$$I = 1/a$$

Substituting for l in equation (3)

$$\frac{1}{a - x} = k t + \frac{1}{a}$$

$$k t = \frac{1}{a - x} - \frac{1}{a}$$

$$k = \frac{1}{t} \cdot \frac{x}{a(a - x)}$$

Characteristics of second order reaction

1) Half life of a second order reaction depends on initial concentration.

When $x = a/2$, $t = t_{1/2}$

$$t_{1/2} = \frac{1}{k} \cdot \frac{a/2}{a(a - a/2)}$$

$$t_{1/2} = \frac{1}{k} \cdot \frac{a/2}{a \cdot \frac{a}{2}}$$

$$t_{1/2} = \frac{1}{k} \cdot \frac{a}{2} \times \frac{2}{a \cdot a}$$

$$t_{1/2} = \frac{1}{k} \cdot \frac{1}{a}$$

$$t_{1/2} \propto \frac{1}{a}$$

Methods to Determine order of a reaction

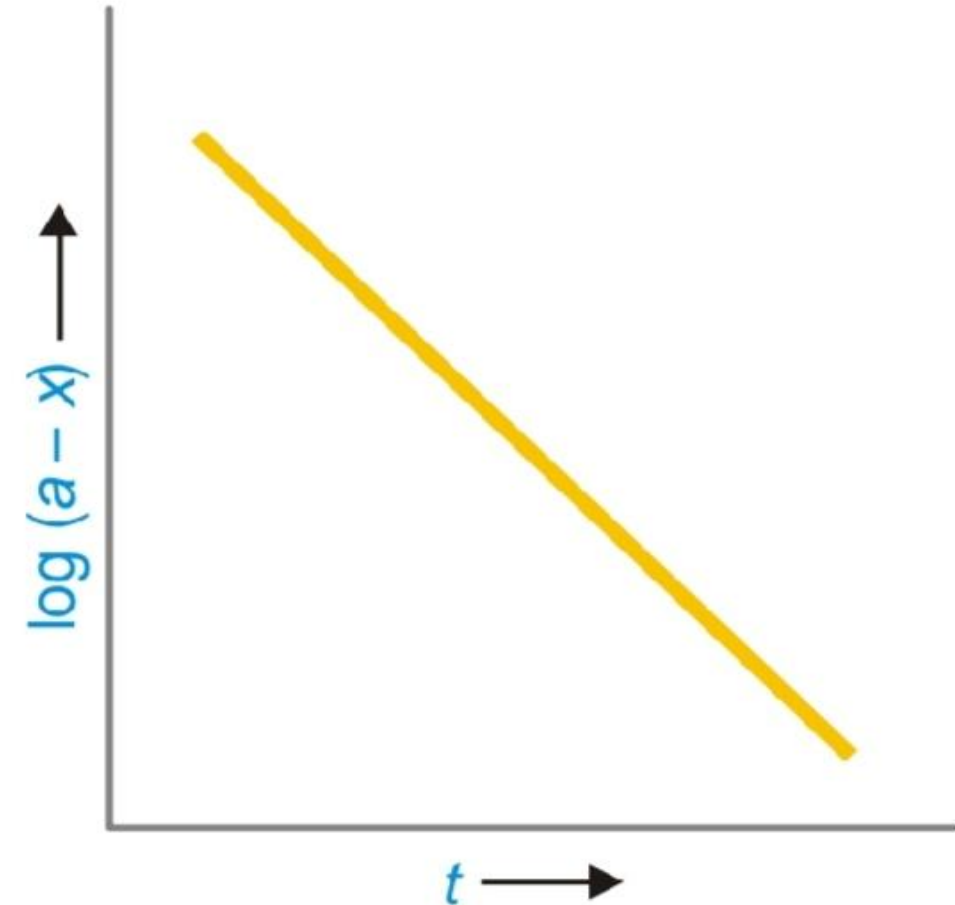
(1) Graphical Method

a) For a first order reaction

$$k = \frac{1}{t} \ln \frac{a}{a - x}$$

$$\ln(a - x) = -kt + \ln a$$

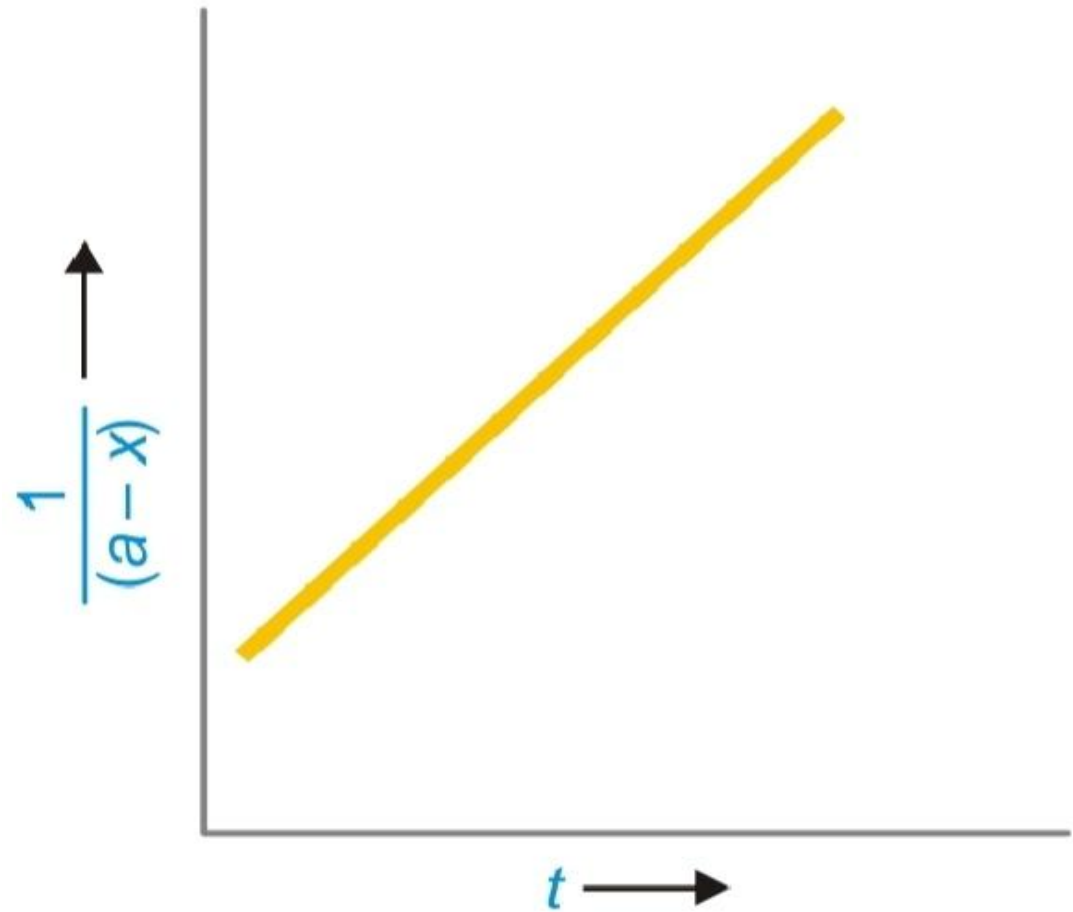
$$\begin{array}{ccccc} \uparrow & & \uparrow & & \uparrow \\ y = & mx + & b \end{array}$$



■ **Figure**
Plot of $\log(a - x)$ against t for a first order reaction.

b) For a second order reaction

$$\begin{array}{ccccc} \frac{1}{a-x} & = & kt & + & \frac{1}{a} \\ \uparrow & & \uparrow & & \uparrow \\ y & = & mx & + & b \end{array}$$



■ **Figure**

Plot of $1/(a - x)$ against t for a second order reaction.

2) Van't Hoff's Differential Method

The method involves determination of rates $(-dc/dt)$ by measuring slopes of concentration (C) vs time (t) curves. For nth order reaction we have,

$$- \frac{dc}{dt} = kc^n$$

For different concentrations we have

$$- \frac{dC_1}{dt} = k C_1^n \quad \dots(1)$$

$$- \frac{dC_2}{dt} = k C_2^n \quad \dots(2)$$

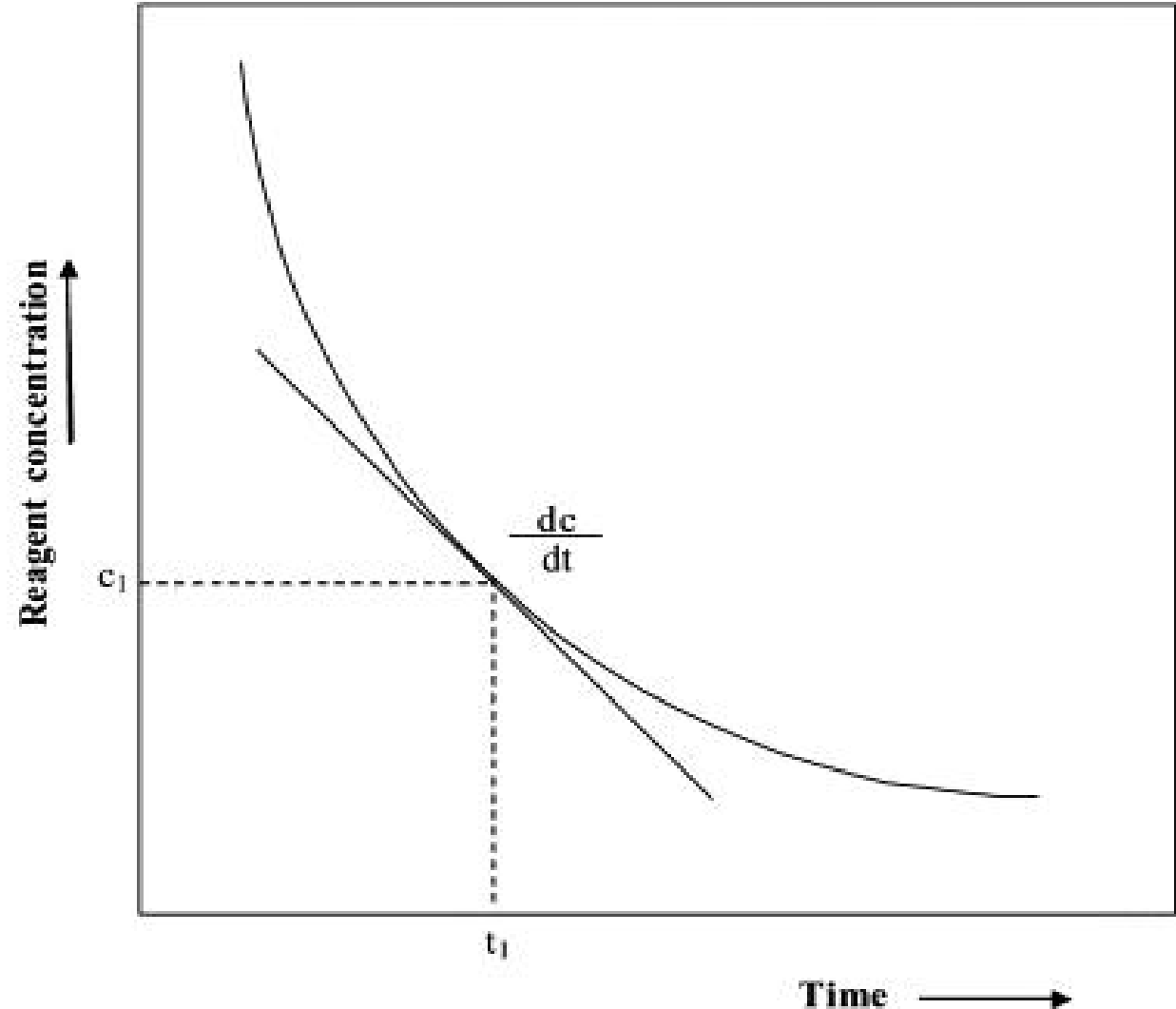
Taking logs of the expression (1) and (2)

$$\log \left(- \frac{dC_1}{dt} \right) = \log k + n \log C_1 \quad \dots(3)$$

$$\log \left(- \frac{dC_2}{dt} \right) = \log k + n \log C_2 \quad \dots(4)$$

On subtracting (4) from (3), we get

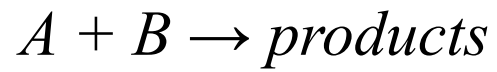
$$n = \frac{\log\left(-\frac{dC_1}{dt}\right) - \left(-\frac{dC_2}{dt}\right)}{\log C_1 - \log C_2}$$



Pseudo First Order Reaction

These reaction which are not such of the first order but under certain condition they becomes of first order.also be called pseudo molecular reactions.

Let us consider a reaction



in which the reactant B is present in a large excess. Since it is an elementary reaction, its rate law can be written as

$$rate = k [A] [B]$$

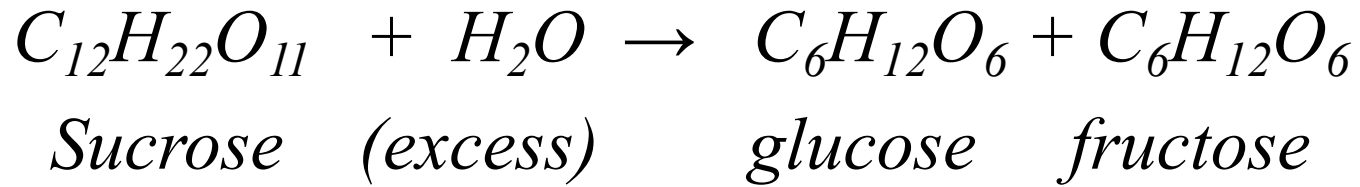
As B is present in large excess, its concentration remains practically constant in the course of reaction. Thus the rate law can be written as

$$rate = k' [A]$$

where the new rate constant $k' = k [B]$. Thus the actual order of the reaction is second-order but in practice it will be first-order. Therefore, the reaction is said to have a pseudo-first order.

Examples of Pseudo-order Reactions

Hydrolysis of sucrose: Sucrose upon hydrolysis in the presence of a dilute mineral acid gives glucose and fructose.



If a large excess of water is present, $[H_2O]$ is practically constant and the rate law may be written

$$\begin{aligned} \text{rate} &= k [C_{12}H_{22}O_{11}] [H_2O] \\ &= k [C_{12}H_{22}O_{11}] \end{aligned}$$

The reaction though of second-order is experimentally found to be first-order. Thus it is a pseudo first-order reaction.

Effect Of Increase Of Temperature On Reaction Rate

The rate of reaction is dependent on temperature. This is expressed in terms of temperature coefficient which is a ratio of two rate constants differing by a temperature of 10°. Generally the temperature selected are 298K and 308K.

The value of temperature coefficient for most of the reactions lies between 2 to 3.

Arrhenius derived a mathematical expression to give a quantitative relationship between rate constant and temperature. The expression is

$$k = A \cdot e^{-E_a / RT} \dots\dots\dots(1)$$

(Here, A = frequency factor; E_a = activation energy ; R = gas constant and T = temperature).

$$\ln k = \ln A + \ln e^{-\left(\frac{E_a}{RT}\right)}$$

$$\ln k = \ln A - \left(\frac{E_a}{RT}\right) \quad (\because \ln e = 1)$$

$$\ln k = \ln A - \left(\frac{E_a}{R}\right)\left(\frac{1}{T}\right) \quad \dots(2)$$

$$y = c + m x$$

A plot of $\ln k$ vs $1/T$ gives a straight line with a negative slope $-E_a/R$. If the rate constant for a reaction at two different temperatures is known, we can calculate the activation energy as follows.

At temperature $T = T_1$; the rate constant $k = k_1$

$$\ln k_1 = \ln A - \left(\frac{E_a}{RT_1}\right) \quad \dots(3)$$

At temperature $T = T_2$; the rate constant $k = k_2$

$$\ln k_2 = \ln A - \left(\frac{E_a}{RT_2} \right) \quad \dots(4)$$

(4) – (3)

$$\ln k_2 - \ln k_1 = - \left(\frac{E_a}{RT_2} \right) + \left(\frac{E_a}{RT_1} \right)$$

$$\ln \left(\frac{k_2}{k_1} \right) = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$2.303 \log \left(\frac{k_2}{k_1} \right) = \frac{E_a}{R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$$

$$\log \left(\frac{k_2}{k_1} \right) = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$$

$$\ln k_2 - \ln k_1 = - \left(\frac{E_a}{RT_2} \right) + \left(\frac{E_a}{RT_1} \right)$$

This equation can be used to calculate from rate constants k_1 and k_2 at temperatures T_1 and T_2 .

Example ;The rate constant of a reaction at 400 and 200K are 0.04 and 0.02 s⁻¹ respectively. Calculate the value of activation energy.

$$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$$

$$T_2 = 400\text{K} ; \quad k_2 = 0.04 \text{ s}^{-1}$$

$$T_1 = 200\text{K} ; \quad k_1 = 0.02 \text{ s}^{-1}$$

$$\log\left(\frac{0.04 \text{ s}^{-1}}{0.02 \text{ s}^{-1}}\right) = \frac{E_a}{2.303 \times 8.314 \text{ JK}^{-1}\text{mol}^{-1}} \left(\frac{400\text{K} - 200\text{K}}{200\text{K} \times 400\text{K}} \right)$$

$$\log(2) = \frac{E_a}{2.303 \times 8.314 \text{ JK}^{-1}\text{mol}^{-1}} \left(\frac{1}{400\text{K}} \right)$$

$$E_a = \log(2) \times 2.303 \times 8.314 \text{ JK}^{-1}\text{mol}^{-1} \times 400\text{K}$$

$$E_a = 2305 \text{ J mol}^{-1}$$

Calculation of E_a Using Arrhenius Equation

Arrhenius Equation

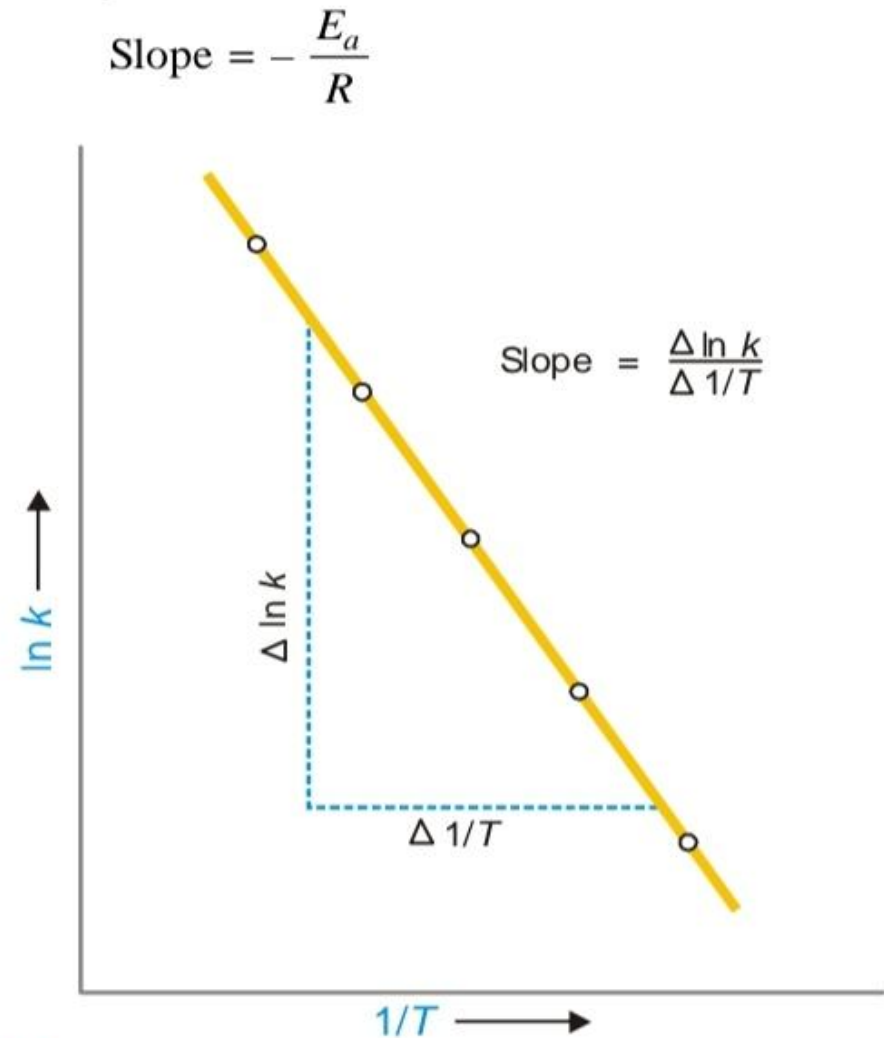
$$k = A \cdot e^{-E_a / RT}$$

$$\ln k = -E_a / RT + \ln A$$

$$\ln k = -\frac{E_a}{R} \left(\frac{1}{T} \right) + \ln A$$



$$y = mx + b$$



■ Figure

The plot of $\ln k$ versus $1/T$ gives a straight line. The slope of line $\Delta \ln k / \Delta 1/T$ gives E_a using the expression given above.

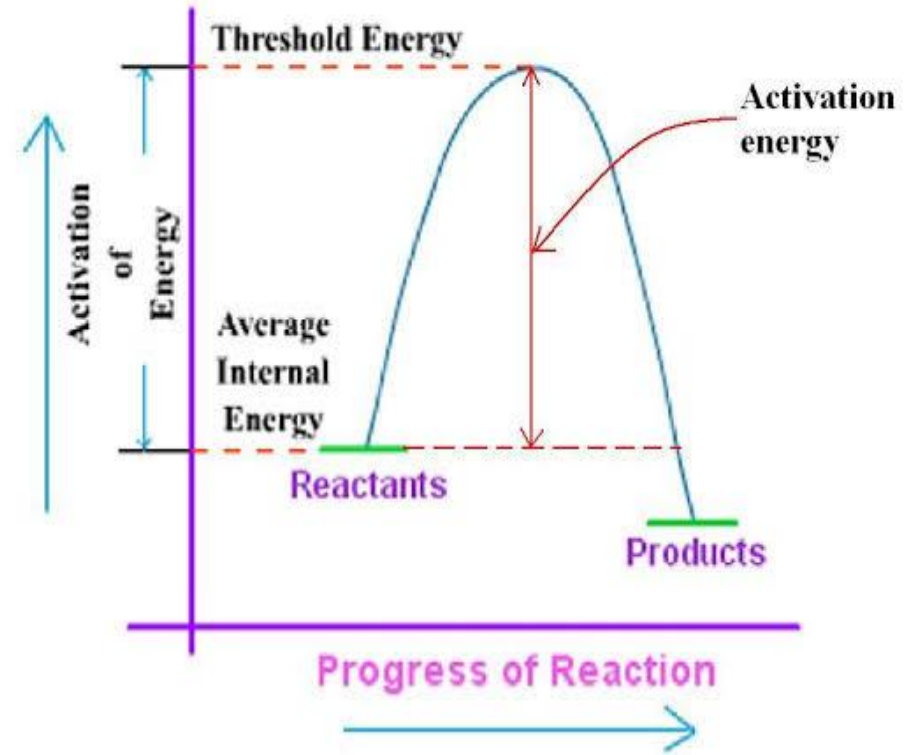
Threshold Energy & Activation Energy

Threshold Energy

For a reaction to take place the reacting molecules must collide together, but only those collisions, in which colliding molecules possess certain minimum energy is called threshold energy .

Activation Energy (E_a) :

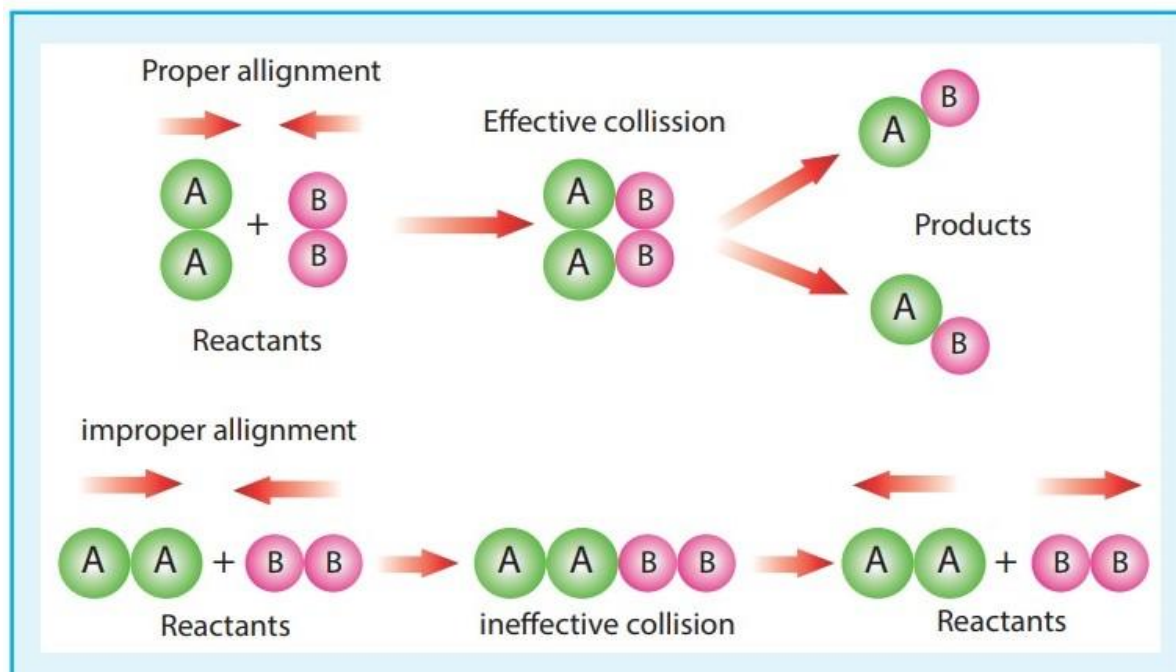
The extra energy needed for the reactant molecules to be able to react chemically is known as Activation energy.



Collision theory of reaction rate

According to this theory, a chemical reaction takes place only by collisions between the reacting molecules.

- ❖ The reacting particles must collide with each other
- ❖ The reacting particles must collide with a certain minimum energy known as the activation energy E_a
- ❖ The reacting particles collide in the correct orientation.



Transition State Theory

The transition state or activated complex theory was developed by Henry Eyring (1935). This theory is also called the absolute rate theory.

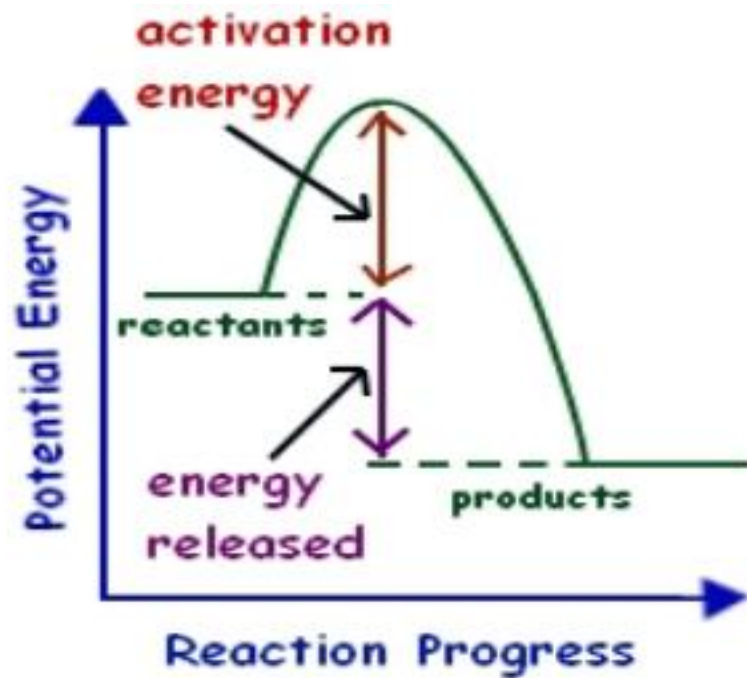
During the collision, the reactant molecules form a transition state or activated complex which decomposes to give the products.



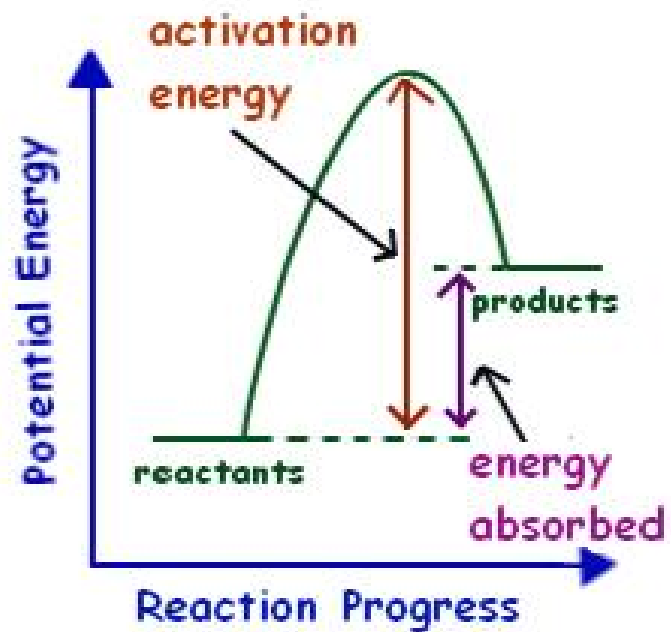
The double dagger superscript ($\ddagger$) is used to identify the activated complex.

The transition state theory may be summarised as follows :

- (1) In a collision, the fast approaching reactant molecules (A and BC) slow down due to gradual repulsion between their electron clouds. In the process the kinetic energy of the two molecules is converted into potential energy.
- (2) As the molecules come close, the interpenetration of their electron clouds occurs which allows the rearrangement of valence electrons.
- (3) A partial bond is formed between the atoms A and B with corresponding weakening of B – C bond. This leads to formation of an activated complex or transition state. The activated complex is momentary and decomposes to give the products (A–B + C)



**Exothermic
reaction**

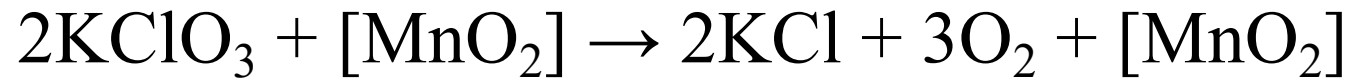


**Endothermic
reaction**

Effect of Catalyst :

A catalyst is a substance, which increases the rate of reaction without itself being consumed at the end of the reaction, and the phenomenon is called catalysis.

Thermal decomposition of KClO_3 is found to be accelerated by the presence of MnO_2 . Here, MnO_2 (foreign substance) act as a catalyst

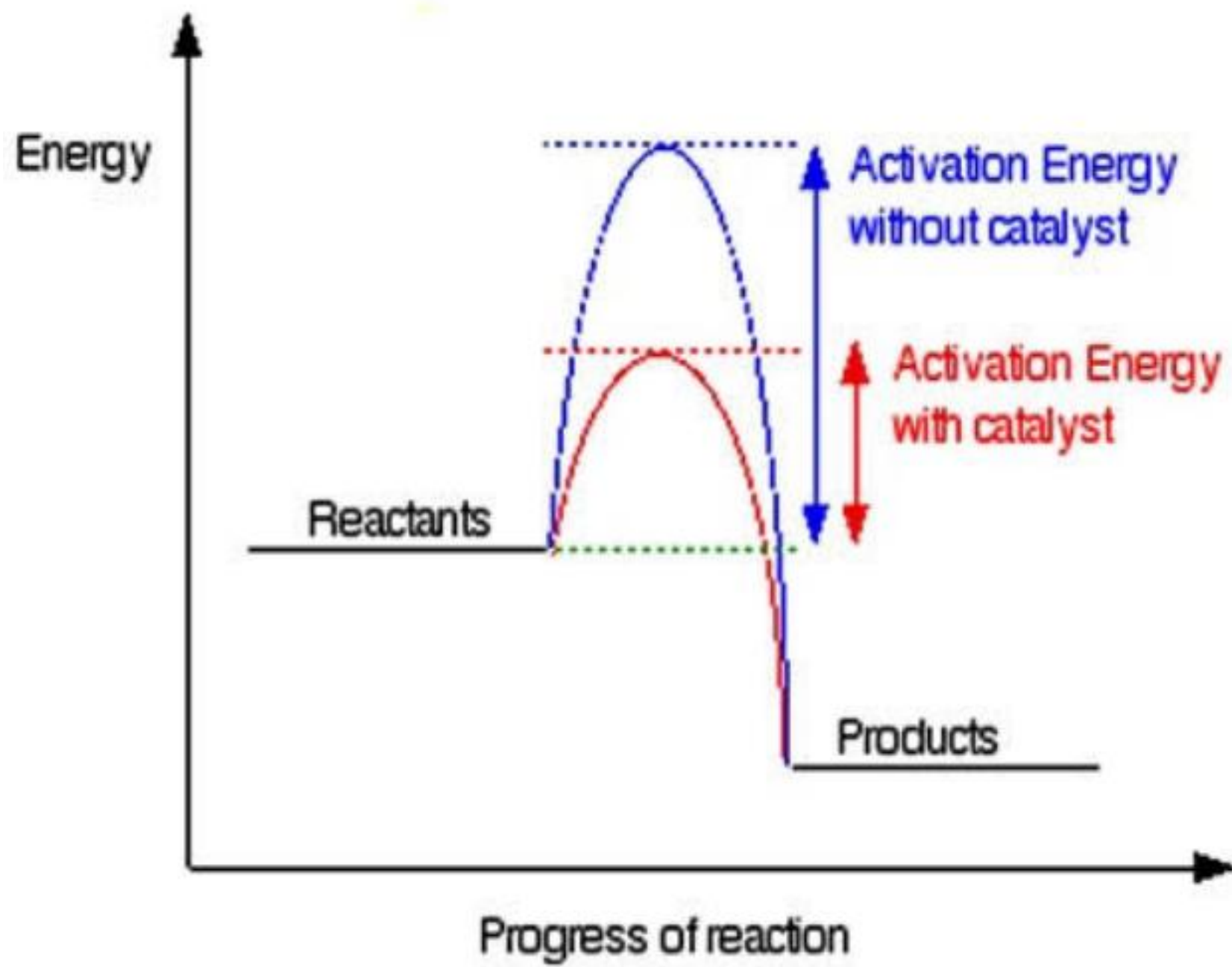


General Characteristic of Catalyst

A catalyst does not initiate the reaction, it simply fastens it.

Only a small amount of catalyst can catalyse the reaction.

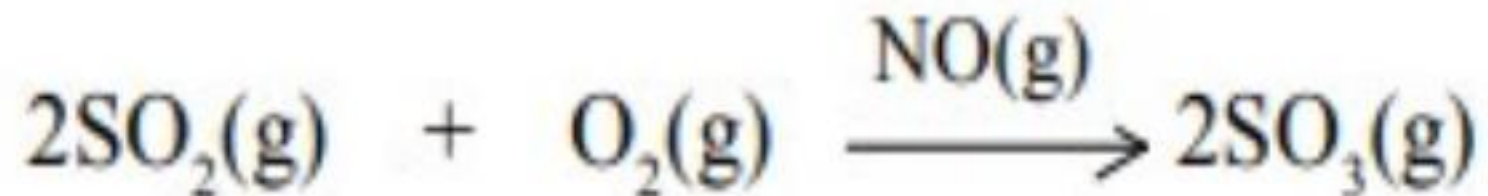
A catalyst drives the reaction through a different route for which energy barrier is of shortest height and Hence, E_a is of lower magnitude. That is, the function of the catalyst is to lower down the activation energy.



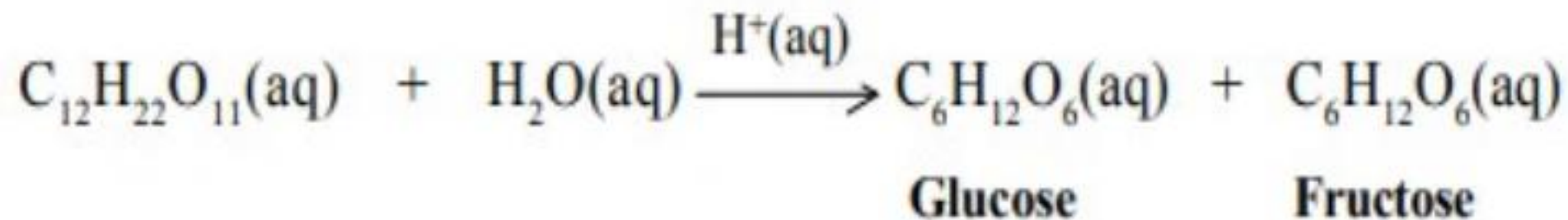
Homogeneous Catalysis

When the catalyst is present in the same phase as the reactants, the phenomenon is called homogeneous catalysis. For example :

(i) Nitric oxide catalyses the oxidation of sulphur dioxide to sulphur trioxide in the lead chamber process.



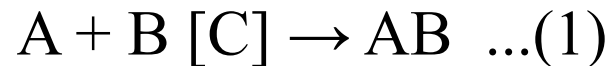
(ii) Hydrogen ions catalyse the inversion of cane sugar



Mechanism of Homogeneous Catalysis

The Intermediate Compound Formation Theory

As already discussed a catalyst functions by providing a new pathway of lower activation energy. In homogeneous catalysis, it does so by forming an intermediate compound with one of the reactants. The highly reactive intermediate compound then reacts with the second reactant to yield the product, releasing the catalyst. Let us illustrate it by taking the general reaction



where C acts as catalyst. The reaction proceeds through the reactions :



Intermediate

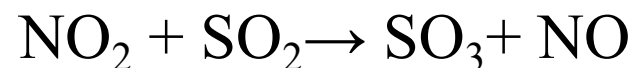


The activation energies of the reactions (2) and (3) are lower than that of the reaction (1) Hence the involvement of the catalyst in the formation of the intermediate compound and its subsequent decomposition, accelerates the rate of the reaction (1) which was originally very slow

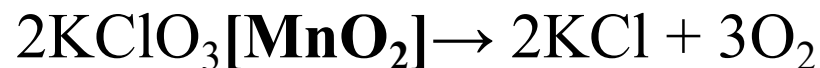
Example 1. Catalytic oxidation of sulphur dioxide (SO₂) in the presence of nitric oxide (NO) as catalyst. (Chamber Process of Sulphuric acid)



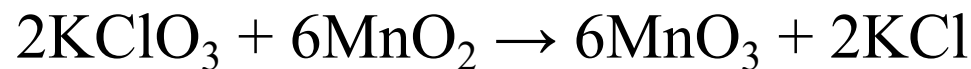
Mechanism :



Example 3. Thermal decomposition of potassium chlorate (KClO₃) in the presence of manganese dioxide (MnO₂).



Mechanism :



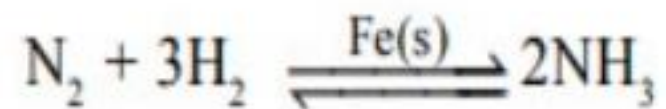
Intermediate
compound



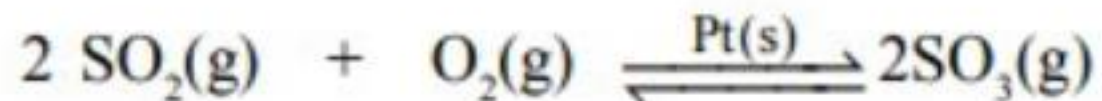
Heterogeneous Catalysis

When the catalyst is present in a phase other than that of reactants the phenomenon is called heterogeneous catalysis. For example :

(i) Iron (s) catalyses the formation of NH_3 gas.



(ii) In contact process for the manufacture of sulphuric acid, platinized asbestos is used as the catalyst



Mechanism of Heterogeneous Catalysis

The Adsorption Theory

This theory explains the mechanism of a reaction between two gases catalysed by a solid (Heterogeneous or Contact Catalysis). Here the catalyst functions by adsorption of the reacting molecules on its surface.

Generally speaking, four steps can be put forward for heterogeneous catalysis. For example, if the reaction is :



Step 1. Adsorption of Reactant molecules

The reactant molecules A and B strike the catalyst surface. They are held up at the surface by weak van der Waals forces (Physical adsorption) or by partial chemical bonds (Chemisorption).

Step 2. Formation of Activated complex

The particles of the reactants adjacent to one another join to form an intermediate complex (A – B). The activated complex is unstable. It has only fleeting existence.

Step 3. Decomposition of Activated complex

The activated complex breaks to form the products C and D. The separated particles of the products hold to the catalyst surface by partial chemical bonds

Step 4. Desorption of Products

The particles of the products are desorbed or released from the surface. They are stable and can lead an independent existence.

The mechanism of contact catalysis may vary in details, depending on the nature of the reactants